

Ashwani Punia, PhD

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RESEARCH PROFILE

As a resourceful and dedicated biochemist, I have over five years of experience in research using mass spectrometry. My passion lies in phytochemistry, metabolomics, and proteomics, and I have published my findings in highly renowned Q1 peer-reviewed journals. My expertise includes a strong background in a variety of chromatographic and spectrometric instruments, such as HPLC (Waters, Agilent, Shimadzu), UPLC (Waters), UPLC-MS/MS (Waters), UHPLC-DAD-QTOF-IMS (Agilent), MALDI-TOF-TOF-MS (Bruker), and GC-MS (Shimadzu). I also possess a solid understanding of various laboratory techniques, including spectrometric assays, separation techniques, ultra-sonication, and both qualitative and quantitative antioxidant assays. Additionally, I am proficient in scientific writing, publishing, and data analysis software like MetaboAnalyst, MS DIAL, XC-MS, MPP, SpectraMill, PAST, GRAPHPAD PRISM, and MeV, and I am skilled in both Linux and Windows operating systems.

RESEARCH INTERESTS

Metabolomics, proteomics and lipidomics using mass spectrometry with chemometric analysis.

Development of novel extraction, isolation and separation techniques for biomolecules.

Analytical Chemistry: Method development and validation.

Separation of isomers using Ion-Mobility Mass Spectrometry (IMS).

Nutri-metabolomics and its application.

Spatial biology using imaging mass spectrometry-DESI-MS, MALDI-MSI and FT-ICR.

Oxidized lipidomics, chronic disease biomarkers, precision health, and nutritional metabolomics.

Clinical cohort studies, translational metabolomics, and mass spectrometry-based multi-omics research.

EDUCATION

Doctor of Philosophy (Biological Sciences) Academy of Scientific & Innovative Research (AcSIR), CSIR-IHBT, Palampur, Himachal Pradesh, India (2020- 2025).

Supervisor: Dr Rajiv Kumar, Principal Scientist, CSIR-IHBT, Palampur, Himachal Pradesh, India

Thesis Title: Development of metabolite reference map and unravelling transcriptional signatures of light dependent aconite biosynthesis in *Aconitum heterophyllum*

Master of Technology (Agricultural Biotechnology), Indian Institute of Technology-Kharagpur, Kharagpur, West Bengal, India (2018-2020).

Master of Science (Biotechnology), Central University of Haryana, Jant-Pali, Haryana, India (2016-2018).

Bachelor of Science (Pass course), Maharshi Dayanand University, Rohtak, India (2012-2016).

RESEARCH EXPERIENCE (2020-till date)

LIST OF PUBLICATIONS with impact factors 2024

1. Thakur, S., Kumari, S., **Punia, A.**, Patial, V., Singh, D. and Kumar, R., 2026. Multi-omics analysis reveals the protective effect of a novel bioactive peptide (BP1) in cardiomyopathy using a zebrafish model. *Journal of Proteome Research*, (IF-3.8)
2. **Punia, A.**, Satyakam., Saini, V., Chouhan, M., Chakraborty, N. and Kumar, R. 2026. Dancing to the tune of the light regime: from perception to secondary metabolite biosynthesis. *Critical Reviews in Biotechnology (IBTY)*, 47, pp 1-21 (IF 8.7)
3. Thakur, S., **Punia, A.**, Acharya, V., Kumar, B., Prasad, A., Yadav, S.K. and Kumar, R., 2024. Bringing bioactive peptides into drug discovery: Challenges and opportunities for medicinal plants. *Industrial Crops and Products*, 222, p.119855. (IF-6.4)
4. **Punia, A.**, Kumari, M., Chouhan, M., Saini, V., Joshi, R., Kumar, A. and Kumar, R., 2024. Proteomic and metabolomic insights into seed germination of *Ferula assa-foetida*. *Journal of Proteomics*, 300, p.105176. (IF-3.1)
5. **Punia, A.**, Joshi, R. and Kumar, R., 2022. Identification and quantification of eight alkaloids in *Aconitum heterophyllum* using UHPLC-DAD-QTOF-IMS: A valuable tool for quality control. *Phytochemical Analysis*, 33(7), pp.1121-1134. (IF-2.9)
6. Kumari, M., Pradhan, U.K., Joshi, R., **Punia, A.**, Shankar, R. and Kumar, R., 2021. In-depth assembly of organ and development dissected *Picrorhiza kurroa* proteome map using mass spectrometry. *BMC Plant Biology*, 21, pp.1-18. (IF-5.6)

Book Chapters

1. Saini, V., **Punia, A.**, Choudhary, A., Vishal, Singh, P.K. and Jayaraman, N.K., 2024. Unlocking Ethnomedicinal Plant Potentials: Advancing Drug Discovery in the High-Throughput Omics Era. **In Ethnomedicinal Plants for Drug Discovery: Current Developments** (pp. 443-462). Singapore: Springer Nature Singapore.
2. **Punia, A.**, Satyakam and Kumar, R., 2025. Mass Spectrometry-Based Methods for Plant Metabolite Profiling. *Plant Stress Memory: Methods and Protocols*, pp.103-117.

Scientific Publications Metrics

Total publications: **8**

Citation: **59+** (Google Scholar)

h-index: **4** i10 index: **3**

Book Chapter: **2**

Paper presented in conferences: **3**

CONFERENCES / POSTER PRESENTATION / ORAL PRESENTATIONS

- “73rd Conference on Mass Spectrometry and Allied Topics” organized by American Society for Mass Spectrometry at Baltimore, Maryland from June 1-5, 2025.
- “8th STUDENT SEMINAR SERIES-2024” held on the occasion of Teachers Day at CSIR-IHBT, Palampur on September 5th, 2024
- “International Symposium on Plant Biology and Functional Genomics (ISPBFG 2024)” held on March 13rd-15th, 2024 conducted by University of Hyderabad, Hyderabad, Telangana India.
- “BIOTECHNOLOGY: EXPLORING THROUGH INNOVATION” held on November 13th - 14th, 2017 organized by Department of Biotechnology, Central University of Haryana, Jant-Pali, Haryana, India.

- **“BIOTECHNOLOGY FOR SUSTAINABLE DEVELOPMENT (BSD-2016)”** held on November 4th-5th, 2016 organized by Department of Biotechnology, Central University of Haryana, Jant-Pali, Haryana, India.

TRAININGS / WORKSHOPS

- **“Mass Spectrometry Workshop”** held on April 3rd – 5th, 2026 organized by MetaChem Academy, India.
- **“DATA REPRODUCIBILITY WORKSHOP SERIES”** held on February 9th – 11th, 2026 organized by Resilience Institute, Michigan State University. United States.
- **“The 2025 Mass Spectrometry Digital Summit”** held on July 22nd and 23rd, 2025 organized by LabX Media Group, Midland, Ontario.
- **“Spatial Metabolomics Workshop”** held on March 5th, 2025 organized by Plant Resilience Institute, Michigan State University. United States.
- **“Research Ethics, Plagiarism Checking & Removal”** workshop held on February 17th, 2024 conducted by Research Graduate.
- **“The Broad Utility (and Potential Pitfalls) of qPCR in Diagnosing Infectious Diseases”** seminar held on October 25th, 2023 organized by Technology Networks, Sudbury, United Kingdom.
- **“Advances in Proteomics & Metabolomics”** seminar held on October 11th-12th, 2023 organized by Technology Networks, Sudbury, United Kingdom.
- **“IP Awareness/Training program”** held on April 22th, 2022 organized by Intellectual Property Office, India.

MEMBER OF PROFESSIONAL SOCIETIES

- **NORDIC Metabolomics Society**, Copenhagen, Denmark
- **Metabolomics Brazil Portal**, Brazil
- **Metabolomics South Africa**, South Africa